July 28, 2026

Editorial Board
npj Systems Biology and Applications

Dear Editors,

We are pleased to submit our manuscript entitled **"Validated Synthetic Patient Generation from Small Longitudinal Cohorts Using Stochastic Attention"** for consideration as an Article in *npj Systems Biology and Applications*.

Small longitudinal clinical cohorts, common in maternal health and rare disease research, present a fundamental challenge: too few patients to support computational modeling, yet too costly and slow to expand through additional enrollment. In this work, we address this gap with multiplicity-weighted Stochastic Attention (SA), a generative framework rooted in modern Hopfield network theory that produces synthetic patients by interpolating between stored patient profiles on a continuous energy landscape. Unlike parametric methods such as multivariate normal sampling or deep generative models (GANs, VAEs), SA operates directly on the geometry of the observed cohort and requires no parameter estimation, making it well suited to the $n < p$ regime that characterizes most small clinical studies.

We applied SA to a longitudinal coagulation dataset from 23 pregnant patients spanning 72 biochemical features across three visits, including rare subgroups (polycystic ovary syndrome, n=3; preeclampsia, n=5). We validated the synthetic patients through four progressively stringent levels: marginal plausibility, cross-visit covariance preservation, conditional subgroup amplification, and mechanistic consistency with a separately specified, generator-blind 58-ODE model calibrated on the same cohort. Across these validation levels, the synthetic patients closely matched their real counterparts under the reported metrics; non-significant difference tests are treated as descriptive non-detection rather than proof of equivalence. A downstream utility test further showed that a mechanistic model calibrated entirely on synthetic patients predicted held-out real patient outcomes as well as one calibrated on real data.

We believe this work is well suited to *npj Systems Biology and Applications* for several reasons. First, the methodology bridges systems biology (mechanistic ODE modeling of the coagulation cascade) with modern machine learning (Hopfield networks, Langevin dynamics), a combination that aligns with the journal's scope. Second, the multi-level validation framework, particularly the mechanistic validation using a separately specified, generator-blind ODE model calibrated on the same cohort, provides a template that transfers to any domain with a calibrated mechanistic model (pharmacokinetic, physiological, metabolic). Third, the multiplicity-weighted conditional generation capability addresses a practical need in rare disease and maternal health research, where generating condition-specific synthetic cohorts from very small subgroups supports exploratory simulation and power-analysis workflows without increasing the subgroup's effective inferential sample size or replacing recruitment.

A preprint of this work is available on arXiv (arXiv:2604.07557). The manuscript has not been submitted elsewhere.

All code, validation scripts, and generated datasets are publicly available at `https://github.com/varnerlab/SA-generation-legacy-dataset-paper`.

We confirm that all authors have approved the manuscript and that there are no competing interests to declare.

Thank you for considering our submission. We look forward to your response.

Sincerely,

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